Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 13: Input Resistance and the Δ–Y Transform

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Colour code used in these notes and in the slides

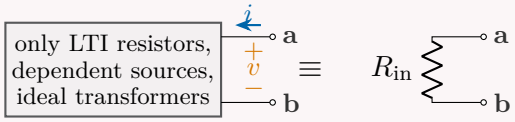
current i voltage v, node voltage e a Δ a Y

Motivation. Previously, we simplified circuits with equivalent one-ports: series and parallel connections and source transformation. Two cases need more. A one-port with dependent sources is not a series–parallel connection of resistors. A bridge has no two resistors in series or in parallel. This lecture finds one equivalent resistor in both cases: the **input resistance** of a one-port, and the **Δ–Y transform** of three resistors.

1. The input resistance of a one-port

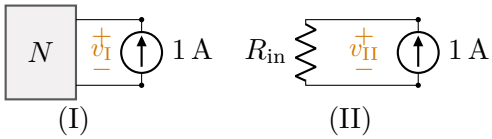
Input resistance

Let a one-port N contain only LTI resistors, LTI dependent sources and ideal transformers, with every control variable inside N. Then N is equivalent to a single LTI resistor R_{in}, the **input resistance** of N: $v = R_{\text{in}}\, i$ at its port.

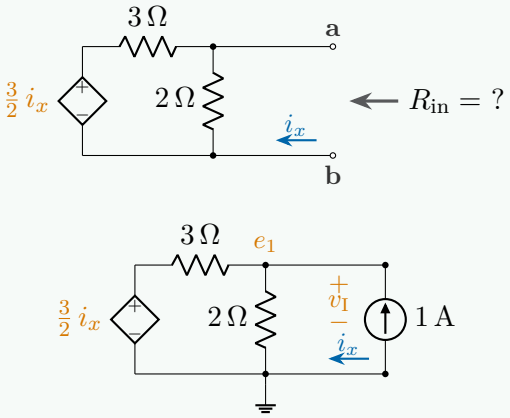


With dependent sources, series and parallel rules are not enough to find R_{in}. A **test source** finds it. Connect a 1 A current source to the port of N (circuit I) and to R_{in} (circuit II). Since $N \equiv R_{\text{in}}$, the port voltages are equal: $v_{\text{I}} = v_{\text{II}} = R_{\text{in}} \cdot 1\,\text{A}$, so

$$R_{\text{in}} = \frac{v_{\text{I}}}{1\,\text{A}}: \quad \text{solve circuit (I).}$$



Example 13.1 — an input resistance



Question. Find R_{in}.

Answer. Connect a 1 A test source and take the bottom node as the ground. KCL at node 1:

$$\frac{e_1 - \frac{3}{2}\, i_x}{3} + \frac{e_1}{2} - 1 = 0.$$

The test current enters at a and leaves at b, so $i_x = -1\,\text{A}$. First plug it in:

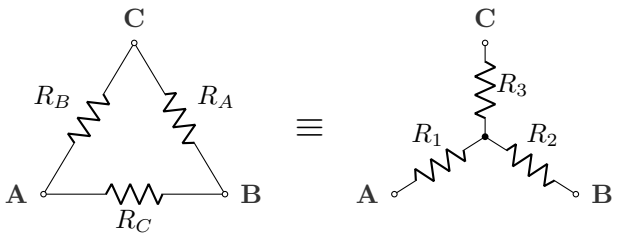
$$\frac{e_1 + \frac{3}{2}}{3} + \frac{e_1}{2} - 1 = 0 \;\Rightarrow\; \frac{5}{6}\, e_1 = \frac{1}{2} \;\Rightarrow\; e_1 = \frac{3}{5}\,\text{V}.$$

So $v_{\text{I}} = e_1 = \frac{3}{5}\,\text{V}$ and $R_{\text{in}} = \frac{3}{5}\,\Omega$.

With the dependent source replaced by a short circuit, R_{in} would be $3 \parallel 2 = \frac{6}{5}\,\Omega$: the dependent source changes the input resistance.

2. The Δ–Y transform

Three resistors may form a Δ or a Y between three terminals A, B, C. In the Δ, R_A is opposite A (between B and C), R_B is opposite B and R_C is opposite C. In the Y, R_1, R_2, R_3 are at A, B, C.



Δ to Y. Given R_A, R_B, R_C, find R_1, R_2, R_3. If the two are equivalent, the resistance between any two terminals, with the third one open, is the same. Between A and B, the Δ is R_C in parallel with $R_A + R_B$, and the Y is $R_1 + R_2$:

$$\frac{R_C(R_A+R_B)}{R_A+R_B+R_C} = R_1+R_2 \quad (1), \qquad \frac{R_A(R_B+R_C)}{R_A+R_B+R_C} = R_2+R_3 \quad (2),$$

$$\frac{R_B(R_C+R_A)}{R_A+R_B+R_C} = R_3+R_1 \quad (3).$$

Half of the sum of (1), (2) and (3) gives

$$R_1+R_2+R_3 = \frac{R_AR_B+R_BR_C+R_CR_A}{R_A+R_B+R_C} \qquad (4),$$

and (4)−(2), (4)−(3), (4)−(1) give the result.

Δ to Y

$$R_1 = \frac{R_BR_C}{R_A+R_B+R_C}, \qquad R_2 = \frac{R_CR_A}{R_A+R_B+R_C}, \qquad R_3 = \frac{R_AR_B}{R_A+R_B+R_C}$$

The Y resistor at a terminal is the product of the two Δ resistors at that terminal, divided by the sum of the Δ resistors.

Y to Δ. Given R_1, R_2, R_3, find R_A, R_B, R_C. From the result above,

$$S = R_1R_2+R_2R_3+R_3R_1 = \frac{R_AR_BR_C\,(R_A+R_B+R_C)}{(R_A+R_B+R_C)^2} = \frac{R_AR_BR_C}{R_A+R_B+R_C},$$

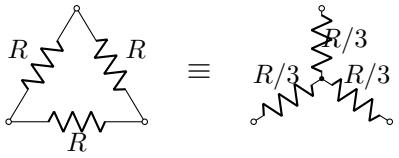
$$R_AR_1 = R_BR_2 = R_CR_3 = \frac{R_AR_BR_C}{R_A+R_B+R_C} = S.$$

Y to Δ

$$R_A = \frac{S}{R_1}, \qquad R_B = \frac{S}{R_2}, \qquad R_C = \frac{S}{R_3}, \qquad S = R_1R_2+R_2R_3+R_3R_1$$

A Δ resistor is the sum of the products of the Y resistors, divided by the Y resistor opposite to it.

Equal resistors. A Δ of three equal resistors R is equivalent to a Y of three resistors $R \cdot R/(3R) = R/3$. Conversely, a Y of three resistors R is equivalent to a Δ of three resistors $3R$.

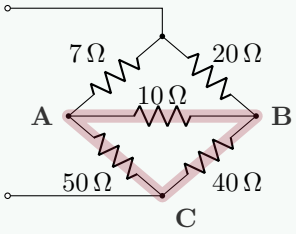


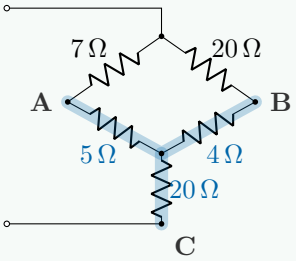
Exercise. Write both transforms with conductances, $G_\square = 1/R_\square$. *Answer:* Y to Δ becomes $G_A = G_2G_3/(G_1+G_2+G_3)$, and so on; it has the form of Δ to Y with resistances.

3. An example: a bridge

Example 13.2 — a bridge

Question. Find the resistance R between the two terminals. No two resistors are in series or in parallel.





Answer. Transform the Δ A–B–C into a Y: $R_C = 10\,\Omega$, $R_B = 50\,\Omega$, $R_A = 40\,\Omega$, and $R_A + R_B + R_C = 100\,\Omega$. So

$$R_1 = \frac{50 \cdot 10}{100} = 5\,\Omega, \qquad R_2 = \frac{10 \cdot 40}{100} = 4\,\Omega, \qquad R_3 = \frac{40 \cdot 50}{100} = 20\,\Omega.$$

In the new circuit, $7 + 5 = 12\,\Omega$ and $20 + 4 = 24\,\Omega$ are in parallel, $12 \parallel 24 = 8\,\Omega$, and this is in series with $20\,\Omega$:

$$R = 8 + 20 = 28\,\Omega.$$

Check. With node voltages, a 1 A test source between the terminals gives a terminal voltage of 28 V. ✓

Summary

In the circuit	What to do
a one-port of resistors and dependent sources	connect a 1 A test source; $R_{\text{in}} = v/1\,\text{A}$
a Δ	Y: $R_1 = R_B R_C/(R_A + R_B + R_C)$, $R_2 = R_C R_A/(R_A + R_B + R_C)$, $R_3 = R_A R_B/(R_A + R_B + R_C)$
a Y	Δ: $R_A = (R_1 R_2 + R_2 R_3 + R_3 R_1)/R_1$, and so on
equal resistors	Δ with $R \equiv$ Y with $R/3$